2) Data Mapping

Dataset Overview

This study uses the EPIC data tables from the Medical Informatics Operating Room Vitals and Events Repository (MOVER), which contains electronic medical record data on surgeries performed at the University of California, Irvine (UCI) Medical Center from 2017 to 2022, comprising records for 39,685 patients and 64,354 surgeries [7].

Identifiers and unit of analysis

The dataset includes two identifiers: MRN (a patient-level identifier) and LOG_ID (an encounter-level identifier). A single patient (MRN) may have multiple surgical encounters, resulting in multiple LOG_ID values. Because postoperative hypoxemia and perioperative variables are encounter-specific, the analytic unit for this study is the surgical encounter, represented by LOG_ID.

Data scope and construct mapping

We will use three EPIC tables from the MOVER EPIC dataset archive (EPIC_EMR.tar.gz): Patient Information (patient_information.csv), Patient Labs (patient_labs.csv), and Patient Postoperative Complications (patient_post_op_complications.csv). These tables collectively provide encounter-level demographics, perioperative timestamps, preoperative laboratory test results, and the postoperative complications used to define the hypoxemia outcome. For preoperative laboratory predictors, we consider a targeted set of lab tests motivated by prior studies (i.e., leukocytes, pH, hematocrit, CRP, and lactate), along with the five most frequently observed lab tests in the dataset beyond the targeted ones (carbon dioxide, sodium, glucose, hemoglobin, and potassium).

Variable mapping

Table 1 summarizes how each construct is represented in the available data tables, including variable names and measurement types.

Construct	Table	Columns	Level	Type
Postoperative hypoxemia (outcome)	Patient Postoperative Complications	SMRTDTA_ELEM_VALUE	Encounter	Character
Demographics (age, sex, BMI)	Patient Information	BIRTH_DATE, SEX, HEIGHT, WEIGHT	Encounter	Numeric, Categorical, Numeric, Numeric
Lab test result (leukocytes, pH, hematocrit, CRP, lactate, carbon dioxide, sodium, glucose, hemoglobin, potassium)	Patient Labs	Lab Code, Lab Name, Observation Value, Measurement Units, Collection Datetime	Encounter-lab record	Character, Character, Numeric, Character, Datetime

Surgery start time (anesthesia timing)	Patient Information	AN_START_DATETIME	Encounter-event record	Datetime

Table 1. Construct-to-variable mapping (encounter key: LOG_ID).

Note: Collection Datetime (Patient Labs) and AN_START_DATETIME (Patient Information) are alignment timestamps used to relate laboratory measurements to the surgical timeline (e.g., identifying the most recent test results before anesthesia start). These fields will not be included as predictors in the final analytic table.

Together, these mapped fields support constructing an encounter-level dataset indexed by LOG_ID, containing the hypoxemia outcome, demographics (age/sex/BMI), and preoperative laboratory measurements (leukocytes, pH, hematocrit, CRP, lactate, carbon dioxide, sodium, glucose, hemoglobin, and potassium).

References

[7] Muntaha Samad, Mirana Angel, Joseph Rinehart, Yuzo Kanomata, Pierre Baldi, and Maxime Cannesson. 2023. Medical Informatics Operating Room Vitals and Events Repository (MOVER): a public-access operating room database. *JAMIA Open* 6, 4 (2023), ooad084. https://doi.org/10.1093/jamiaopen/ooad084